

PCCST501 – COMPUTER NETWORKS

Assignment 3 | Module 3 – Data-Link Layer, Wireless LANs & Mobile IP

LAN, Ethernet & Wireless Network Investigation

Name	Muhammed Aamir
Roll No.	48
Course	PCCST501 – Computer Networks
Programme	B.Tech Computer Science & Engineering
College	Viswajyothi College of Engineering and Technology
Course Outcome	CO3
Due Date	17 August 2026

Assignment Overview

This report answers all ten questions in the supplied Assignment 3 question sheet, covering the data-link layer, Ethernet, switching, CSMA/CD, CSMA/CA, wireless LAN design, Mobile IP, and wired-versus-wireless frame comparison.

Submission note: The Wireshark sections contain clearly marked spaces for the student's own screenshots and captured MAC-address values. These cannot be truthfully completed without an actual capture.

Question 1: Data-link layer responsibilities; LLC vs MAC

The data-link layer is responsible for transferring frames between directly connected nodes over a local link. It takes packets from the network layer and organizes them into frames that can be delivered across the physical network.

- **Framing:** Encapsulates a network-layer packet into a frame and identifies its boundaries.
- **MAC addressing:** Uses hardware addresses to identify the sender and receiver on the local link.
- **Error detection:** Uses mechanisms such as the Frame Check Sequence (FCS/CRC) to detect corrupted frames.
- **Flow control:** Can prevent a fast sender from overwhelming a slower receiver.
- **Medium access control:** Determines when a station is allowed to transmit on a shared medium.

LLC and MAC sub-layers

Feature	LLC – IEEE 802.2	MAC – IEEE 802.3 / 802.11
Position	Upper part of Layer 2; closer to the network layer.	Lower part of Layer 2; interfaces with the physical medium.
Main role	Provides a logical interface and can identify/multiplex upper-layer protocols.	Handles MAC addressing, framing details, FCS and access to the medium.
Medium dependence	More independent of the physical transmission technology.	Specific to the LAN technology being used.
Example	Identifying the network protocol carried in the payload.	Ethernet frame construction or Wi-Fi channel access.

Summary: LLC provides the logical link-control functions, while MAC deals with the actual local medium, addressing and frame transmission.

Question 2: Ethernet frame format

An Ethernet II frame carries an upper-layer packet across an Ethernet link. The main fields and their functions are shown below.

Preamble	SFD	Destination MAC	Source MAC	Type/Length	Payload + Pad	FCS
7 B	1 B	6 B	6 B	2 B	46–1500 B	4 B

Figure 1: Ethernet II frame – field sizes and order

Field	Size	Purpose
Preamble	7 bytes	Clock synchronization pattern before the frame.
SFD	1 byte	Marks the start of the actual Ethernet frame.
Destination MAC	6 bytes	Hardware address of the intended receiver; may be unicast, multicast or broadcast.
Source MAC	6 bytes	Hardware address of the transmitting interface.
Type / Length	2 bytes	EtherType identifies the upper-layer protocol; in 802.3 framing the field can indicate payload length.
Payload + Pad	46–1500 bytes	Carries upper-layer data. Padding is added when necessary to meet the minimum frame size.
FCS	4 bytes	CRC-based error-detection value used to detect corruption.

Important sizes: the normal Ethernet frame has a minimum size of 64 bytes from Destination MAC through FCS and a maximum of 1518 bytes without an 802.1Q VLAN tag. The payload MTU is normally 1500 bytes.

Question 3: Wireshark capture – source and destination MAC addresses

The supplied assignment requires at least five Ethernet frames to be captured in Wireshark. The actual MAC addresses should be taken from the student's own capture rather than invented in the report.

- **Start capture:** open Wireshark and select the active Ethernet/Wi-Fi interface.
- **Generate traffic:** ping a reachable host, open a webpage, or generate ARP traffic.
- **Stop and filter:** stop the capture after sufficient packets have been collected; filters such as eth, arp or icmp can help.
- **Inspect Ethernet II:** select a frame and expand the Ethernet II section to read Source and Destination MAC addresses.
- **Record five frames:** enter the packet number, protocol, source MAC, destination MAC and EtherType.

Screenshot 1 – Wireshark capture

[INSERT YOUR WIRESHARK SCREENSHOT HERE]

Frame	Protocol	Source MAC	Destination MAC	EtherType
1	_____	_____	_____	_____
2	_____	_____	_____	_____
3	_____	_____	_____	_____
4	_____	_____	_____	_____
5	_____	_____	_____	_____

Observation: on a typical LAN, an Internet-bound frame uses the MAC address of the local default gateway as its destination. ARP requests commonly use the broadcast destination FF:FF:FF:FF:FF:FF.

Question 4: Switch operation and MAC address learning

A switch is a Layer-2 device that forwards Ethernet frames using a MAC address table. The table is learned automatically from the source addresses of incoming frames.

- **Learn:** when a frame enters a port, the switch associates the source MAC address with that ingress port.
- **Forward:** if the destination MAC is already in the table, the frame is sent only through the corresponding port.
- **Flood:** if the destination is unknown, or the frame is broadcast, the switch sends it out through the appropriate other ports.
- **Filter:** if the destination is known to be on the same port on which the frame arrived, the switch does not forward it back.

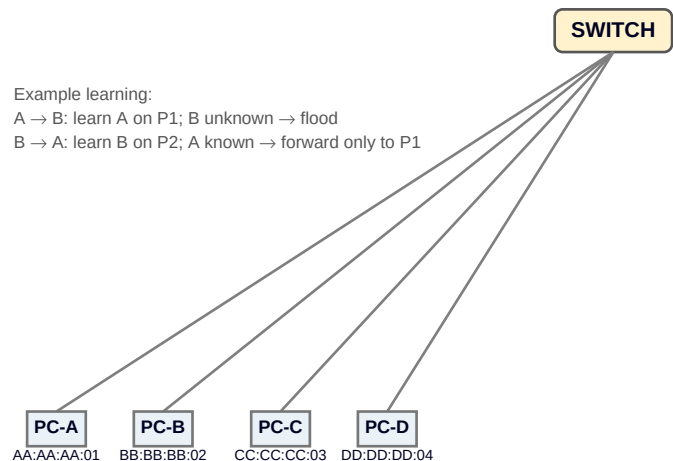


Figure 2: Example of MAC learning with four hosts

Step	Event	Resulting MAC-table idea
0	Switch starts	Table is empty.
1	A sends to B	Learn A on Port 1; B is unknown, so the frame is flooded.
2	B replies to A	Learn B on Port 2; A is known, so forward only to Port 1.
3	C sends to A	Learn C on Port 3; destination A is known.
4	D sends to B	Learn D on Port 4; destination B is known.
5	A sends to C	Both addresses are known; forward only to C's port.

Question 5: Hub vs bridge vs switch

Parameter	Hub	Bridge	Switch
OSI layer	Layer 1	Layer 2	Layer 2 (some advanced models can also perform Layer 3 functions)
Forwarding	Repeats bits to all ports.	Uses MAC addresses between segments.	Uses a MAC table to forward per port.
Collision domain	One shared collision domain.	Separated by bridge ports/segments.	Normally one collision domain per port; full duplex removes collisions.
Bandwidth	Shared by connected hosts.	Shared within each segment.	Dedicated link bandwidth per port.
Practical use	Mostly obsolete.	Older method of connecting LAN segments.	Standard device in modern switched LANs.

Question 6: CSMA/CD and its reduced role in modern Ethernet

CSMA/CD stands for Carrier Sense Multiple Access with Collision Detection. It was used by shared, half-duplex Ethernet networks to coordinate stations using the same medium.

- **Carrier sense:** a station listens to determine whether the medium is idle.
- **Multiple access:** several stations share the same communication medium.
- **Collision detection:** a transmitting station monitors the medium and detects a collision.
- **Jam signal:** after a collision, a jam signal ensures the other stations know that a collision occurred.
- **Binary exponential back-off:** the station waits for a randomly selected delay before attempting retransmission.

Why CSMA/CD is less important today: modern Ethernet normally uses switches and full-duplex point-to-point links. A switched full-duplex link has separate transmit and receive paths, so collisions do not occur and CSMA/CD is not needed. It remains mainly relevant to legacy shared-media or half-duplex Ethernet.

Question 7: CSMA/CA in IEEE 802.11 and comparison with CSMA/CD

Wi-Fi uses CSMA/CA because a wireless station cannot reliably detect a collision while it is transmitting. Hidden terminals can also cause two stations that cannot hear each other to transmit toward the same access point.

- **Carrier sensing:** the station checks whether the wireless channel is busy.
- **IFS waiting:** when the channel is idle, the station waits for the appropriate inter-frame space.
- **Random back-off:** a contention counter is selected and decreases while the channel remains idle.
- **Transmission and ACK:** after a unicast transmission, the receiver sends an ACK; lack of an ACK causes retransmission procedures.
- **RTS/CTS:** optional control frames can reduce hidden-terminal problems by reserving the medium.

Aspect	CSMA/CD	CSMA/CA
Main idea	Detect a collision after it occurs.	Try to avoid collisions before transmission.
Typical medium	Shared/half-duplex Ethernet.	Wireless LAN.
Collision detection	Possible on the shared wired medium.	Not reliably possible during transmission.
ACK at MAC layer	Not normally used for Ethernet data frames.	Used for unicast Wi-Fi frames.
Back-off	After collision.	Before transmission and after failures.
Hidden terminal	Not a typical wired problem.	Important; RTS/CTS and NAV can help.

Question 8: Wireless LAN design for a college laboratory

For a laboratory of approximately 12 m × 8 m with around 40 workstations and up to about 60 concurrent devices, a small multi-AP design provides better capacity and coverage than relying on a single access point.

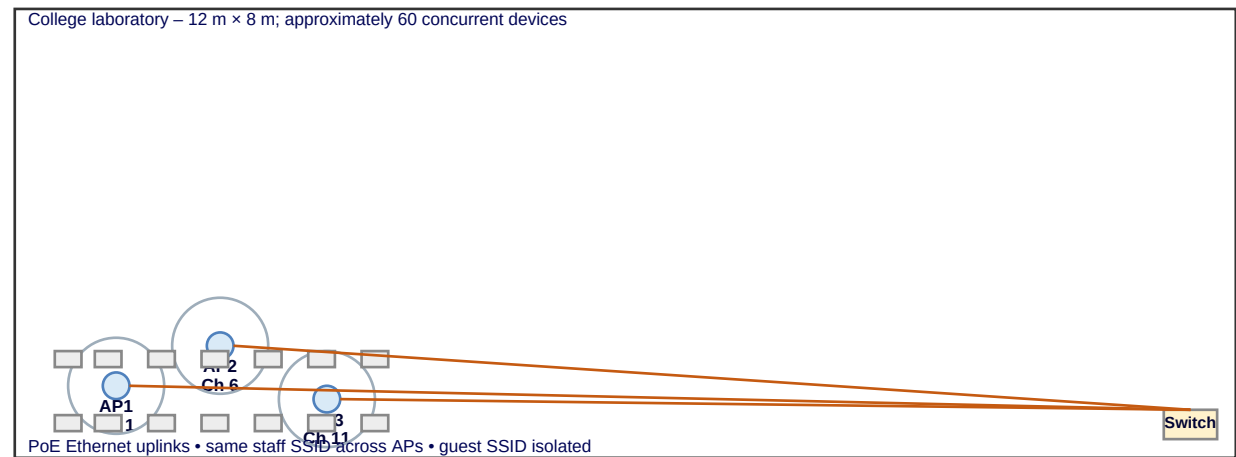


Figure 3: Proposed three-AP laboratory WLAN layout

Design item	Proposed specification
Access points	Three ceiling-mounted dual-band APs, positioned for overlapping coverage and smoother roaming.
Wireless standard	Wi-Fi 5 (802.11ac) or Wi-Fi 6 (802.11ax); prefer 5 GHz where supported.
SSID	College-Lab for staff/students and College-Guest for isolated Internet-only access.
2.4 GHz channels	Use non-overlapping channels 1, 6 and 11 for AP1, AP2 and AP3.
5 GHz channels	Use separated 20/40 MHz channels according to the local RF environment.
IP addressing	Example lab subnet 192.168.10.0/24 with DHCP; a separate guest subnet/VLAN such as 192.168.20.0/24.
Backbone	Gigabit PoE switch connecting APs to a router/firewall.
Security	Prefer WPA3 or WPA2-AES; use strong credentials or 802.1X/RADIUS; disable WEP and WPS.
Capacity	Aim for roughly 20 clients per AP in this example and monitor channel utilization.

The APs should be mounted away from large metal objects and thick obstructions. A small amount of cell overlap is useful for roaming, while careful channel planning reduces co-channel interference.

Question 9: Mobile IP and the need for mobility support

An IP address normally represents both a host and its location through the network prefix. When a mobile device moves to a different network, simply keeping the old address can cause routing problems. Mobile IP allows the device to retain a permanent home address while using a temporary care-of address at the visited network.



Figure 4: Simplified Mobile IPv4 packet delivery

Main components

- **Mobile Node (MN):** the device that changes its point of attachment.
- **Home Agent (HA):** a router on the home network that records the MN's current location and helps deliver packets.
- **Foreign Agent (FA):** a router on a visited network that can provide/assist with a care-of address in Mobile IPv4.
- **Home address:** the permanent address associated with the MN.
- **Care-of Address (CoA):** a temporary address representing the MN's current location.

Operation

- **Agent discovery:** the mobile node learns that it has moved and identifies mobility support.
- **Registration:** the current care-of address is registered with the home agent.
- **Tunnelling:** packets sent to the home address can be intercepted by the HA and encapsulated toward the care-of address.
- **Delivery at the visited network:** the tunnel endpoint decapsulates the packet and delivers it to the mobile node.
- **Reply path:** packets from the mobile node can be sent toward the correspondent node, with reverse tunnelling used when required by network filtering.

Need for mobility support: phones, laptops and other mobile devices regularly move between Wi-Fi networks, cellular networks, campuses and home networks. Mobility mechanisms aim to maintain communication sessions during these changes. Triangle routing can be inefficient; route optimization is an important improvement in later mobility designs.

Question 10: Packet-level comparison of wired Ethernet and wireless LAN frames

Ethernet and IEEE 802.11 both operate at the data-link layer, but their frame structures reflect the different characteristics of wired and wireless media. A Wi-Fi frame contains more control and addressing information because the wireless medium is shared and less predictable.

Feature	Wired Ethernet II	Wireless LAN (802.11)
MAC header	Destination + Source MAC and Type/Length.	Frame Control, Duration/ID, multiple address fields and Sequence Control.
Address fields	Normally two MAC addresses.	Can use up to four MAC address fields depending on To-DS/From-DS.
Frame types	Ethernet data frames at the MAC level.	Management, control and data frames.
Acknowledgement	No normal MAC ACK for each Ethernet data frame.	Unicast data frames use link-layer ACKs.
Virtual carrier sensing	Not used in the same way.	Duration/ID contributes to NAV operation.
Sequence information	Not a standard Ethernet II field.	Sequence Control supports ordering, duplicate detection and fragmentation functions.
Payload	Normally up to 1500 bytes for standard Ethernet MTU.	802.11 frame body can be larger than a standard Ethernet payload depending on frame format and aggregation.
Security	Provided by higher/adjacent mechanisms such as 802.1X/MACsec where deployed.	WPA2/WPA3 provide link-layer wireless protection.

Suggested evidence for the report

Evidence	What to insert
Screenshot 2 – Ethernet	Your Wireshark Ethernet II frame with the Ethernet header expanded.
Screenshot 3 – 802.11	A Wireshark 802.11 capture in monitor mode, or a clearly cited documented example if monitor-mode capture is unavailable.

Conclusion: Ethernet uses a comparatively simple frame structure suited to switched, reliable point-to-point links. IEEE 802.11 adds control, acknowledgement, sequence and additional addressing fields to manage a shared wireless medium.

Deliverables Checklist

Deliverable	Included in this report
Ethernet frame diagram	Question 2 – Figure 1
Wireshark screenshots	Question 3 and Question 10 – insert your own captures
Wireless LAN design	Question 8 – Figure 3 and design table
Comparison tables	Questions 5, 7 and 10
README.md	Prepare/attach in the GitHub repository as required by the assignment

References

- Forouzan, B. A., *Data Communications and Networking*, McGraw-Hill.
- Tanenbaum, A. S. and Wetherall, D., *Computer Networks*, Pearson.
- IEEE 802.3 – Ethernet standards.
- IEEE 802.11 – Wireless LAN standards.
- RFC 5944 – IP Mobility Support for IPv4.
- Wireshark documentation and user guide.

Important: Before submission, fill in your roll number and replace the Wireshark placeholders with readable screenshots and five real frame entries from your own capture.